CHAPTER 5

WEB DESIGNING USING HTML

CHAPTER ESSENCE AT A GLANCE:
HTML provides the semantic structural backbone of the web, organizing multimedia content through standardized markup tags, hyperlinked documents, structured tabular grids, interactive forms, and cascading style rules.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Document Structure and Semantic Tags:** HTML documents follow a standardized structural tree consisting of doctype declaration, root html element, head metadata container, and body display container, utilizing paired container tags and standalone empty tags.

• **Content Structuring: Lists and Tabular Layouts:** Lists organize content into ordered numerical, unordered bulleted, or descriptive definition sequences, while table tags organize multidimensional data using rows, headers, data cells, and multi-span attributes.

• **Hyperlinks and Resource Referencing:** Anchor tags establish hyperlinked navigation across the web using href attributes with absolute or relative Uniform Resource Locators, linking images, documents, and target page bookmarks.

• **Interactive Forms and CSS Presentation:** Form elements capture user input via diverse controls like text fields, radio buttons, checkboxes, and submit triggers, styled and formatted through inline, internal, or external Cascading Style Sheets.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{HTML5 Document Architecture} = \frac{<!\text{DOCTYPE html}> \rightarrow <html>}{<head>(\text{Title, Meta, CSS})</head> + <body>(\text{Content})</body>}$$

HTML Document Tree Stratification

$$\text{Table Cell Spanning} = \frac{\text{colspan} = k \quad (\text{Merge } k \text{ Adjacent Columns Horizontally})}{\text{rowspan} = m \quad (\text{Merge } m \text{ Adjacent Rows Vertically})}$$

Tabular Multi-Cell Span Geometry

$$\text{Form Submission Pipeline} = \frac{\text{GET Method: Appends Parameters to URL Query String (Limited)}}{\text{POST Method: Transmits Parameters Securely in HTTP Body (Bulk)}}$$

HTTP Form Transmission Protocol

$$\text{CSS Hierarchy Cascade} = \text{Inline Style Attribute} > \text{Internal } <style> > \text{Block} > \text{External Linked .css File}$$

Style Sheet Specificity and Precedence

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Always use the POST method rather than GET when submitting forms containing sensitive data like passwords, as GET displays data in the browser URL address bar.

• **Alert:** Radio buttons in a mutually exclusive group must share the exact same name attribute value; giving them different names allows multiple buttons to be checked simultaneously.

• **Alert:** Empty tags such as br, hr, img, and input have no closing tags; never write closing tags like </br> or </img> in HTML documents.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between container tags and empty tags in HTML with two examples of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Container Tags: Require both an opening tag and a corresponding closing tag to enclose content between them (e.g. **...**, <title>...</title>). **[1 Mark]**

• Empty Tags: Standalone tags that do not enclose content and have no closing tag (e.g.
, <hr>, <img>). **[1 Mark]**

2. Distinguish between ordered lists (<ol>) and unordered lists (<ul>), stating two attributes of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Ordered Lists (<ol>): Displays items in numbered sequential order using numbers or letters. Attributes: type ('1', 'A', 'a', 'I', 'i') and start (starting numeric offset). **[1 Mark]**

• Unordered Lists (<ul>): Displays items in unranked bulleted format. Attributes: type ('disc', 'circle', 'square'). **[1 Mark]**

3. Explain colspan and rowspan attributes of <td> tags with a sample HTML code snippet and explain their visual effect. [Score: 3 Marks]

Answer & Key Valuation Steps:

• colspan Attribute: Merges multiple adjacent cells horizontally across columns (e.g. <td colspan='2'> spans two columns). **[1 Mark]**

• rowspan Attribute: Merges multiple adjacent cells vertically across rows (e.g. <td rowspan='3'> spans three rows down). **[1 Mark]**

• Code Illustration: <table border='1'><tr><th colspan='2'>Result</th></tr><tr><td>Subject</td><td>Marks</td></tr></table>. **[1 Mark]**

4. Compare GET and POST methods used in the <form> tag for transmitting user data to the web server. [Score: 3 Marks]

Answer & Key Valuation Steps:

• GET Method: Appends form data directly to the URL string after a question mark (query string); visible in address bar, bookmarks allowed, limited to approximately 2048 characters, unsuitable for passwords. **[1.5 Marks]**

• POST Method: Embeds form data inside the body of the HTTP request; invisible in URL, highly secure for passwords, supports large data transfers including file uploads. **[1.5 Marks]**

5. Write HTML code to create a student registration form containing text input for Name, radio buttons for Gender, checkboxes for Hobbies, and a Submit button. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Form and Name Input: <form action='submit.php' method='POST'> Name: <input type='text' name='userName'>
 [1 Mark]

• Radio and Checkbox Inputs: Gender: <input type='radio' name='gender' value='M'> Male <input type='radio' name='gender' value='F'> Female
 Hobbies: <input type='checkbox' name='hob' value='Read'> Reading <input type='checkbox' name='hob' value='Code'> Coding
 [1 Mark]

• Submit Trigger: <input type='submit' value='Register'></form> **[1 Mark]**

6. Explain the three methods of inserting Cascading Style Sheets (CSS) into an HTML document: Inline, Internal, and External. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Inline Style: Applied directly to a specific HTML tag using the style attribute (e.g. <h1 style='color:blue;'>Heading</h1>); lowest reusability. **[1 Mark]**

• Internal Style Sheet: Embedded within the <head> section using the <style> tag, applying styles across that single HTML page. **[1 Mark]**

• External Style Sheet: Written in a separate .css file and linked to HTML documents using the <link> tag in the head, providing maximum site-wide reusability. **[1 Mark]**

7. Write complete, standards-compliant HTML code to design an educational website homepage featuring a styled navigation bar, an image with alt text, an ordered list of subjects, and a data table. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Document Envelope and Head: <!DOCTYPE html><html><head><title>Mission School</title><style>nav{background:#1E1B4B;padding:10px;}nav a{color:white;margin:10px;text-decoration:none;}</style></head> **[1 Mark]**

• Navigation and Image: <body><nav>HomeCourses</nav><h1>Welcome</h1><img src='school.jpg' alt='School Campus' width='400'>
 [1 Mark]

• Ordered List: <h3>Key Subjects:</h3>
Computer Applications
Accountancy
Economics **[1 Mark]**

• Data Table and Closure: <table border='1'><tr><th>Course</th><th>Duration</th></tr><tr><td>Commerce</td><td>2 Years</td></tr></table></body></html> **[1 Mark]**